When $\cos 5\theta = 0 \Rightarrow 5\theta = (2n+1)\frac{\pi}{2}$ $\Rightarrow \theta = (2n+1)\frac{\pi}{10}, n \in \mathbb{Z}$

When, $\sin 4\theta = 0 \Rightarrow 4\theta = n\pi$ $\Rightarrow \theta = \frac{n\pi}{4}, n \in \mathbb{Z}$

Thus, the general solution of the given equation is $\theta = (2n+1)\frac{\pi}{10}$, $\theta = \frac{n\pi}{4}, n \in \mathbb{Z}$.

Example 3.49 Solve $\tan 2x = -\cot\left(x + \frac{\pi}{3}\right)$

Solution:

$$\tan 2x = -\cot\left(x + \frac{\pi}{3}\right) = \tan\left(\frac{\pi}{2} + x + \frac{\pi}{3}\right) = \tan\left(\frac{5\pi}{6} + x\right)$$
$$2x = n\pi + \frac{5\pi}{6} + x, n \in \mathbb{Z} \Rightarrow x = n\pi + \frac{5\pi}{6}, n \in \mathbb{Z}.$$

Example 3.50 Solve $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$

Solution:

$\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$
$\sin 3x + \sin x - 3\sin 2x = \cos 3x + \cos x - 3\cos 2x$
$2\sin 2x\cos x - 3\sin 2x = 2\cos 2x\cos x - 3\cos 2x$
$(\sin 2x - \cos 2x)(2\cos x - 3) = 0$
Then, $\sin 2x - \cos 2x = 0$ since $2\cos x - 3 \neq 0$
$\sin 2x = \cos 2x \Rightarrow \tan 2x = 1 \Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}, n \in \mathbb{Z}.$

Example 3.51 Solve $\sin x + \cos x = 1 + \sin x\cos x$

Solution:

Let $\sin x + \cos x = t$
$\Rightarrow 1 + 2\sin x\cos x = t^2 \Rightarrow \sin x\cos x = \frac{t^2 - 1}{2}$

Thus, the given equation becomes $t^2 - 2t + 1 = 0 \Rightarrow t = 1$

Hence, $\sin x + \cos x = 1$
$\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x\right) = 1$
$\sqrt{2}\cos\left(\frac{\pi}{4} - x\right) = 1 \Rightarrow \cos\left(\frac{\pi}{4} - x\right) = \frac{1}{\sqrt{2}}$
Thus, we have $x - \frac{\pi}{4} = \pm\frac{\pi}{4} + 2n\pi, n \in \mathbb{Z}$
$x = \frac{\pi}{2} + 2n\pi$, or $x = 2n\pi$, $n \in \mathbb{Z}$.